

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE CODE: CSA5115

COURSE NAME: Cryptography and Network Security

COURSE OUTCOME COVERED

CO3: *Examine cryptographic hash algorithms, digital signature schemes, and assess Kerberos and X.509 authentication frameworks for ensuring integrity, authentication, and non-repudiation.CO (BL4 , BL5)*

Assessment Tool Weightage: 20%

QUESTIONS: 5

TOTAL MARKS: 25

Annexure A

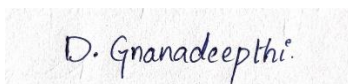
CASE STUDY

Code of Conduct:

I D. Gnana Deepthi (192411123) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Annexure A

Case Study

1. A software company is using MD5 to store user passwords in its database. After a security breach, it is discovered that attackers were able to crack multiple passwords. Analyze the weaknesses of MD5 in this scenario and propose a more secure hashing approach using SHA-based techniques.
2. A banking application implements digital signatures for transaction verification, but users report unauthorized transactions being approved. Analyze the possible flaws in the digital signature implementation and recommend improvements based on secure signature standards.
3. An enterprise network uses Kerberos for authentication, but attackers successfully perform a replay attack to gain unauthorized access. Examine how this attack could occur and propose enhancements in the authentication process to prevent such threats.
4. A secure communication system uses a simple hash function to verify message integrity, but attackers are able to modify messages without detection. Analyze why this approach fails and propose a solution using HMAC to improve message authentication.
5. An organization implements X.509 certificates for secure communication, but users encounter frequent trust validation errors. Analyze the possible issues in certificate management and propose solutions to ensure proper authentication and trust chain validation.

ANSWERS

3. Replay Attack in Kerberos Authentication

Kerberos is a network authentication protocol that uses tickets and secret keys to verify users and services. However, attackers may perform a replay attack by capturing a valid authentication message and sending it again to gain unauthorized access.

How the Attack Occurs

1. The attacker intercepts a valid Kerberos ticket or authenticator sent by a user.
2. The attacker captures the encrypted authentication information from the network.
3. The captured message is replayed to the target server within its validity period.
4. If the server does not properly detect duplicate requests, it may accept the replayed message as a valid authentication request.
5. The attacker may then gain unauthorized access by impersonating the legitimate user.

Enhancements to Prevent Replay Attacks

1. Use Timestamps:

Each Kerberos authenticator should contain a timestamp. The server accepts it only within a small time interval, preventing old messages from being reused.

2. Maintain a Replay Cache:

The server should store recently received authenticators and reject duplicate authentication requests.

3. Use Short-Lived Tickets:

Reducing the ticket validity period limits the time available for attackers to replay captured credentials.

4. Use Nonces:

A unique random number can be included in each authentication session to ensure that every request is unique.

5. Mutual Authentication:

Both the client and server should verify each other's identity, reducing the possibility of impersonation.

Conclusion:

Replay attacks in Kerberos occur when attackers capture and reuse valid authentication messages. Using timestamps, replay caches, nonces, short ticket lifetimes, and mutual authentication can effectively strengthen the authentication process and prevent unauthorized access.

4. Analysis of Simple Hash Function and HMAC Solution

A simple hash function generates a fixed-size hash value for a message to verify its integrity. However, using only a hash function is not secure because an attacker can modify the original message and calculate a new hash value. The attacker can then send both the modified message and its new hash, making the receiver believe that the message is genuine.

Why Simple Hash Fails

1. No secret key: Anyone can calculate the hash of a modified message.
2. No authentication: A hash verifies data integrity but does not verify the sender's identity.
3. Hash replacement: Attackers can replace both the message and its hash value.

Solution Using HMAC

HMAC (Hash-based Message Authentication Code) combines a cryptographic hash function with a secret key shared between the sender and receiver.

Working:

1. The sender generates an HMAC using the message and a secret key.
2. The sender sends the message along with the HMAC value.
3. The receiver uses the same secret key to generate a new HMAC.

4. If both HMAC values match, the message is accepted; otherwise, it is rejected.

Conclusion

Using HMAC improves security because an attacker cannot generate a valid authentication code without knowing the secret key. Thus, HMAC provides both message integrity and authentication, making it more secure than a simple hash function.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand